

EPIC 1bit2bit – Project Document

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Table of Contents

Project Overview	3
Tech stack used	3
Computer Networks & Cybersecurity	4
Network Architecture Diagram	4
TLS Handshake & Certificate Verification	4
Authentication & Authorization	4
Penetration Testing Report	4
C++ Programming.....	5
C++ Component Structure.....	5
Class Diagrams	5
Build & Run Instructions	5
Memory Management & Object Ownership	5
Cryptography	6
Threat Model	6
Cryptographic Primitives and Parameter Justification	6
Construction Walkthrough Diagrams.....	6
Known Limitations	6
Blockchain.....	7
Trust Model & Decentralisation	7
Message Recording.....	7
Message Verification	7
AI Usage Declaration & Reflection.....	8

Table of Figures

Project Overview

Group name: 1bit2qbit

Project URL: 1bit2qbit.theburkenator.com

GitHub Repository: <https://github.com/meltedelement/EPIC-1bit2qbit/>

Name + student number	% contribution	Project work undertaken
Aaron O'Doherty 24424048	33%	• Item • Item • Item • Item • Item • Item
James Hayes 24402796	33%	• Item • Item • Item • Item • Item • Item
Skye Fitzpatrick 24402095	33%	• Item • Item • Item • Item • Item • Item

Tech stack used

Explain, explain, explain

Computer Networks & Cybersecurity

Network Architecture Diagram

TLS Handshake & Certificate Verification

Authentication & Authorization

Penetration Testing Report

C++ Programming

C++ Component Structure

Class Diagrams

Build & Run Instructions

Memory Management & Object Ownership

Cryptography

Threat Model

Adversary	Security Properties Held
Passive network attacker	
Active network attacker	
Honest-but-curious server	
Fully compromised server	

Cryptographic Primitives and Parameter Justification

Construction Walkthrough Diagrams

Registration

Key Publication

Message Sending

Message Receiving

Storage at Rest

Known Limitations

Blockchain

Trust Model & Decentralisation

Message Recording

Message Verification

AI Usage Declaration & Reflection

Group AI Usage Approach & Code Review setup

Aaron O'Doherty – prompting strategy, evaluation, & reflection

James Hayes – prompting strategy, evaluation, & reflection

Skye Fitzpatrick – prompting strategy, evaluation, & reflection